

PCA

Main work is to reduce curse of dimensionality

We reduce features through it

Intro

Data ka jo essence hai higher dimension mae it takes that

Analogy – suppose we are having a match toh usme there is a photographer toh jo 3d mae mach chal raha hai, use that photographer tries to capture in 2d in best possible way

Benefits

Faster execution of algos

Visualization

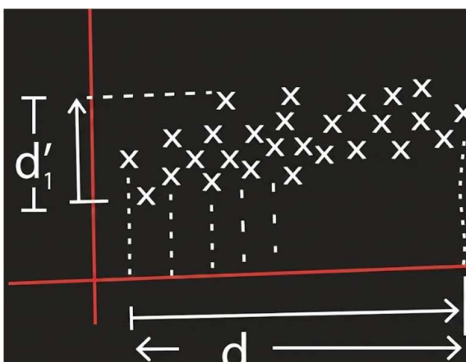
Geometric intuition

No of rooms | no of grocery shops | price

Feature selection – basically unnn features ko u will put jisme se we can get a very good output,

Jab domain knowledge na ho toh –

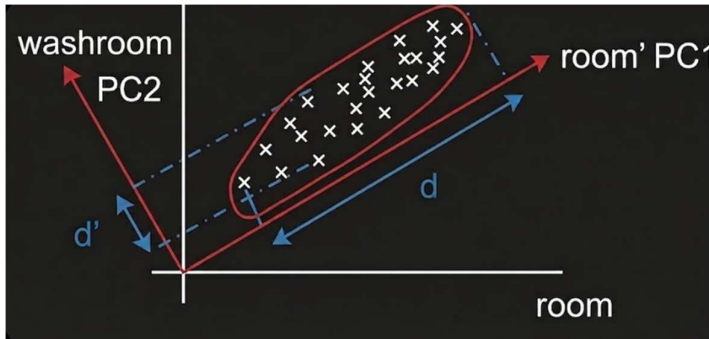
Plot (x = rooms, y = grocery) then data ka spread jiske zyada ho you well keep that



$d > d'$

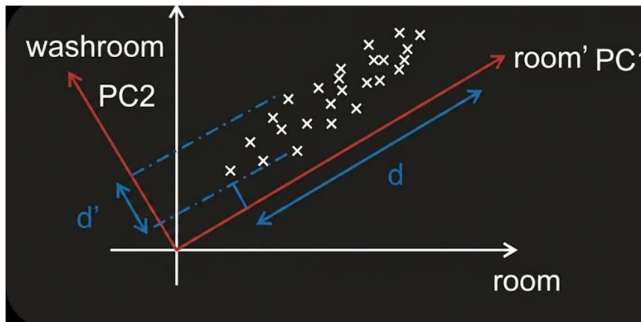
now

no of rooms | no of washrooms | price

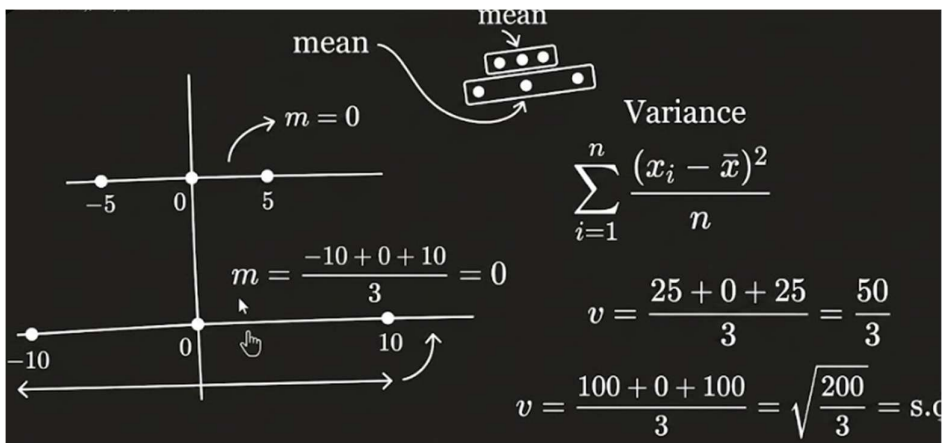


Both are needed

You can do one thing – make a single column 'size' in which rooms ka area and washrooms ka area will be added



Variance



Mean find central term and the variance calculate the distance from the mean point to every point , we get spread ka idea looking at the variance

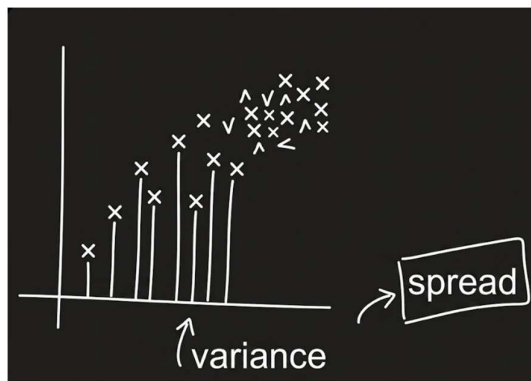
We have 'mean absolute distance' as well jisme we find the distance ka modulus

Now here mod is not differentiable at 0 so we don't use

We get problem in optimal problems ke solution

An **optimization problem** is a problem where you want to find the **best possible solution** from a set of available choices, usually by **maximizing** or **minimizing** some quantity.

Because there is peak yaar to her cannot calculate slope there (sharp point)



this is spread (mean absolute distance)

Variance is so important

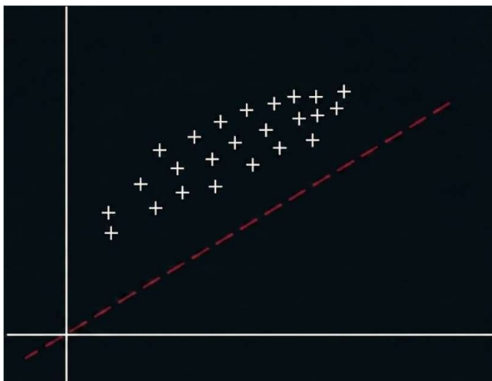
What exactly is happening in pca

Problem Formulation



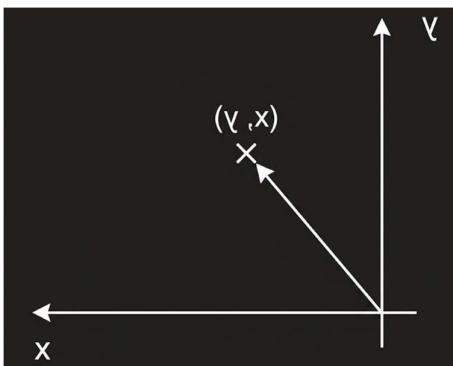
Data set in 2d

Now mujhe ek single axis find karna hai on which I can project my data and should get utna hei acha results as before



It will be that axis

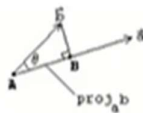
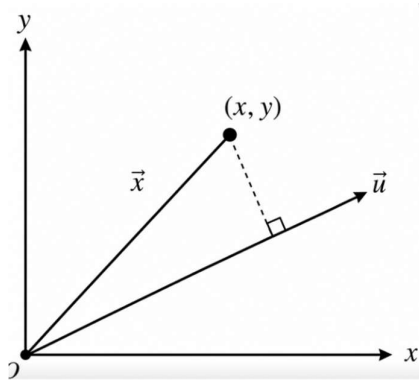
Lets visualize one point (here point is considered as a vector)



aab isse I have to project on another

vector (that red line which was the axis jisme we thought that project kare toh we will get good results)

Find the unit vector for that red axis



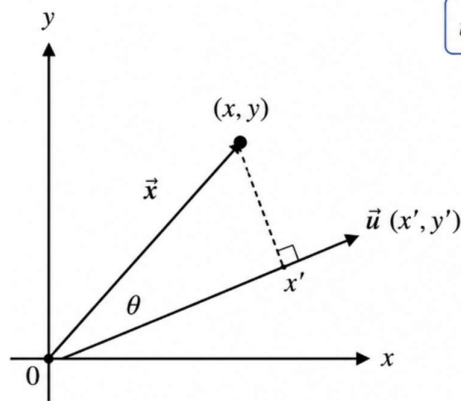
angle between **a** and **b** which from (*) is given by

$$\text{comp}_a \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$$

And, the vector projection is merely the unit vector $\vec{a}/|\vec{a}|$ times the scalar

$$\text{proj}_a \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} \frac{\vec{a}}{|\vec{a}|} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \vec{a}$$

finding the projection



$$\vec{u} \cdot \vec{x} = |\vec{u}| |\vec{x}| \cos \theta = \vec{u}^T \vec{x}$$

$$|\vec{u}| = 1 \quad (\vec{u} \text{ is a unit vector})$$



$$\vec{u} \cdot \vec{x} = \vec{u}^T \vec{x}$$

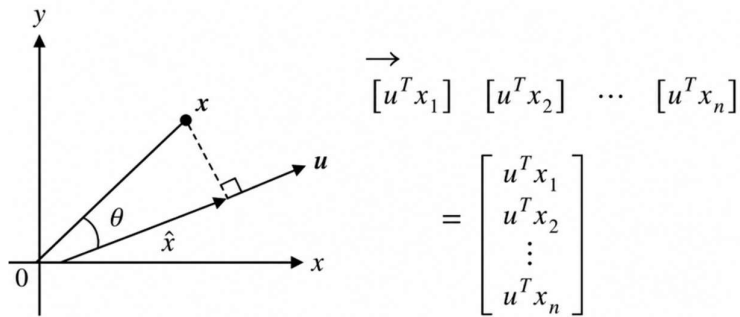
$$\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix} \quad \vec{u} = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$\vec{u}^T = [x' \quad y'] \times \begin{matrix} \text{blue } x \\ \text{red } y \end{matrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$= [x x' + y y'] - \text{scalar}$$

now vector have 2 quantities (x,y), we got a scalar quantity for each vector (only one value)

Which unit vector should we choose – jiska variance is maximum



After projecting all point we will find the variance

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

formula to calculate variance

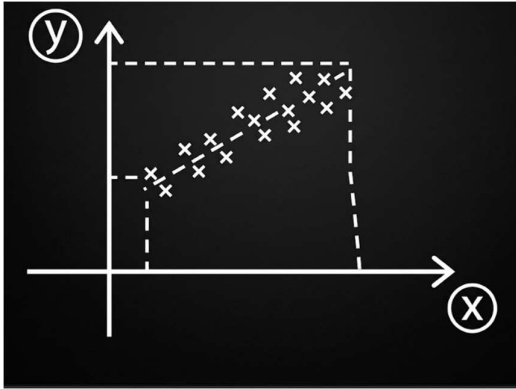
$$\frac{\sum_{i=1}^n \left(u^T x_i - u^T \bar{x} \right)^2}{n}$$

variance on that unit vector (this should be maximum)

Aab yaha pe the $u^T \bar{x}$ is the original data set ka mean point ka projection on that red line

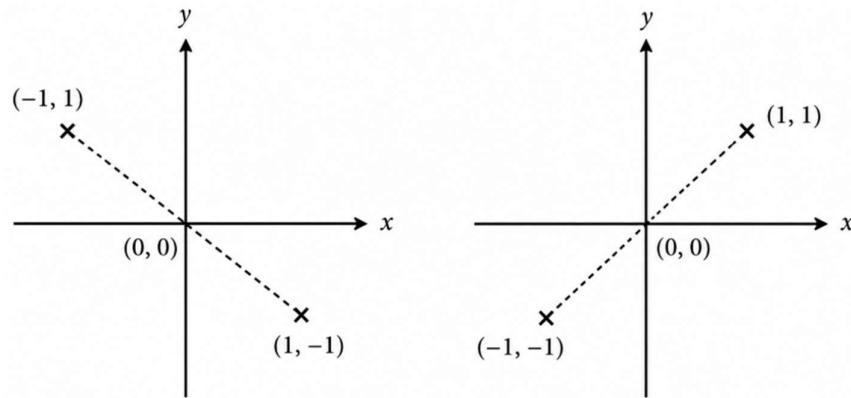
Mean is a measure of central tendency

Problem variance ka –



variance will be different around every axis – x axis pe lag and y axis pe lag

Now it never tells the relationship between x and y



Ye dono data sets ke liye variance will be the same

So , according to variance this both datasets are same

So we launched 'co-variance'

Covariance -

$$Cov(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n}$$

Left Figure

Points:

$$(-1, 1), (1, -1)$$

Mean:

$$\bar{x} = 0, \quad \bar{y} = 0$$

Covariance:

$$\frac{(-1)(1) + (1)(-1)}{2} = \frac{-1 - 1}{2} = -1$$

Right Figure

Points:

$$(-1, -1), (1, 1)$$

Mean:

$$(0, 0)$$

Covariance:

$$\frac{(-1)(-1) + (1)(1)}{2} = \frac{1 + 1}{2} = 1$$

Points Pattern	Covariance
$(-1, -1), (1, 1)$	Positive
$(-1, 1), (1, -1)$	Negative

If no relation then covariance will be zero

Covariance matrix

$$\Sigma = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) \end{bmatrix}$$

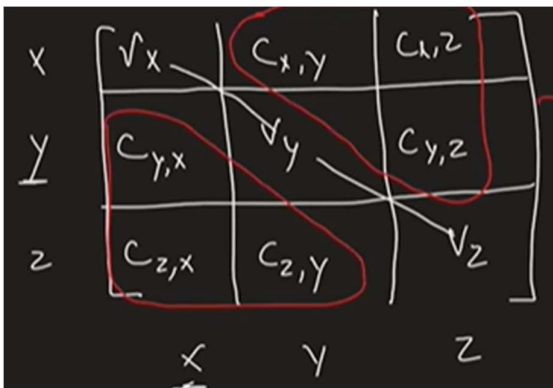
$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_1, X_2) & \text{Var}(X_2) \end{bmatrix}$$

General case (n features)

$$\Sigma = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) & \cdots & \text{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \text{Cov}(X_n, X_2) & \cdots & \text{Cov}(X_n, X_n) \end{bmatrix}$$

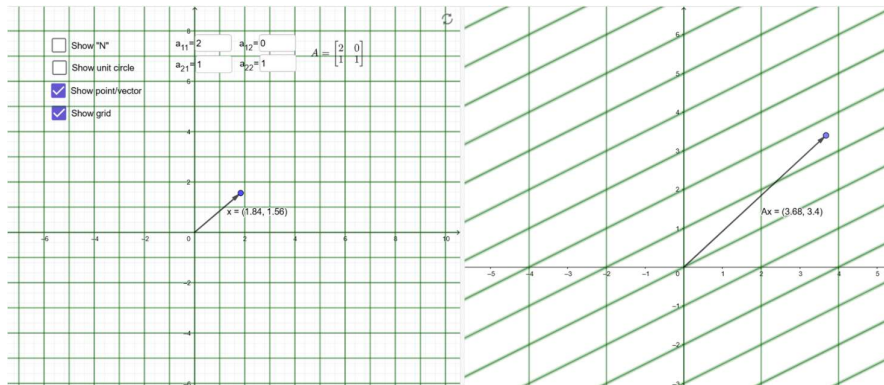
As I learnt before the $\text{cov}(x, y) = \text{cov}(y, x)$

So this matrix will be square and symmetric also



Linear transformation

So basically we have a transformation matrix jiske sath you do apne coordinates ke sath multiplication and u will get another new coordinates cool website - <https://www.geogebra.org/m/YCZa8TAH>



Eigenvector

A vector whose **direction does not change** after a matrix transformation.

Eigenvalue

It tells by how much that eigenvector is stretched or shrunk.

Formula:

$$Av = \lambda v$$

where:

- A = matrix
- v = eigenvector
- λ = eigenvalue

https://www.visiondummy.com/2014/04/geometric-interpretation-covariance-matrix/#google_vignette – a very good blog I found on the internet (studied topic - Eigendecomposition of a covariance matrix)

in simple words isme they told ki you have to first find the covariance matrix and then ussi ko use karke you have to do matrix transformation then we will get some eigen vectors right, from all those eigen vectors jiska bhi eigen value was maximum usika variance was maximum, through this I understood that how we calculate that red line vala unit vector

How pca solves the problem

Suppose we have a 3d data set

F1 | f2 | f3 | target

Aab 3d space ko we have to bring in lower dimension

1 – mean centring (mean = 0)

2 – find covariance matrix

3 – find the eigen values and eigen vectors for the covariance matrix

Because we are working with 3d space – we will get 3 eigen vectors and 3 eigen values($v_1 - pc_1$, $v_2 - pc_2$, $v_3 - pc_3$)

Use Pc_1 and 1d space banao or use pc_1 and pc_2 for 2d dimension